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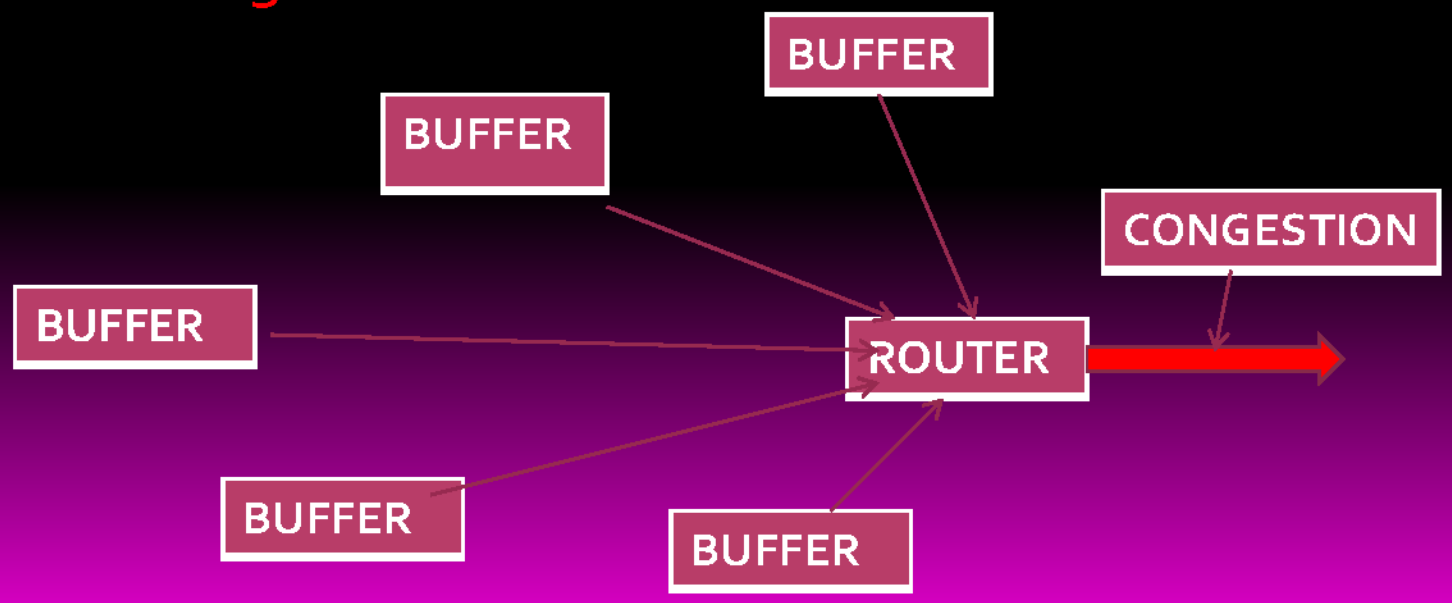
INTRODUCTION

WHAT IS CONGESTION?

Congestion is a situation in communication network in which too many packets are present in a part of the subnet or contending for the same link so

- The queue overflows
- Packets get dropped
- Network is congested!



Factors that Cause Congestion

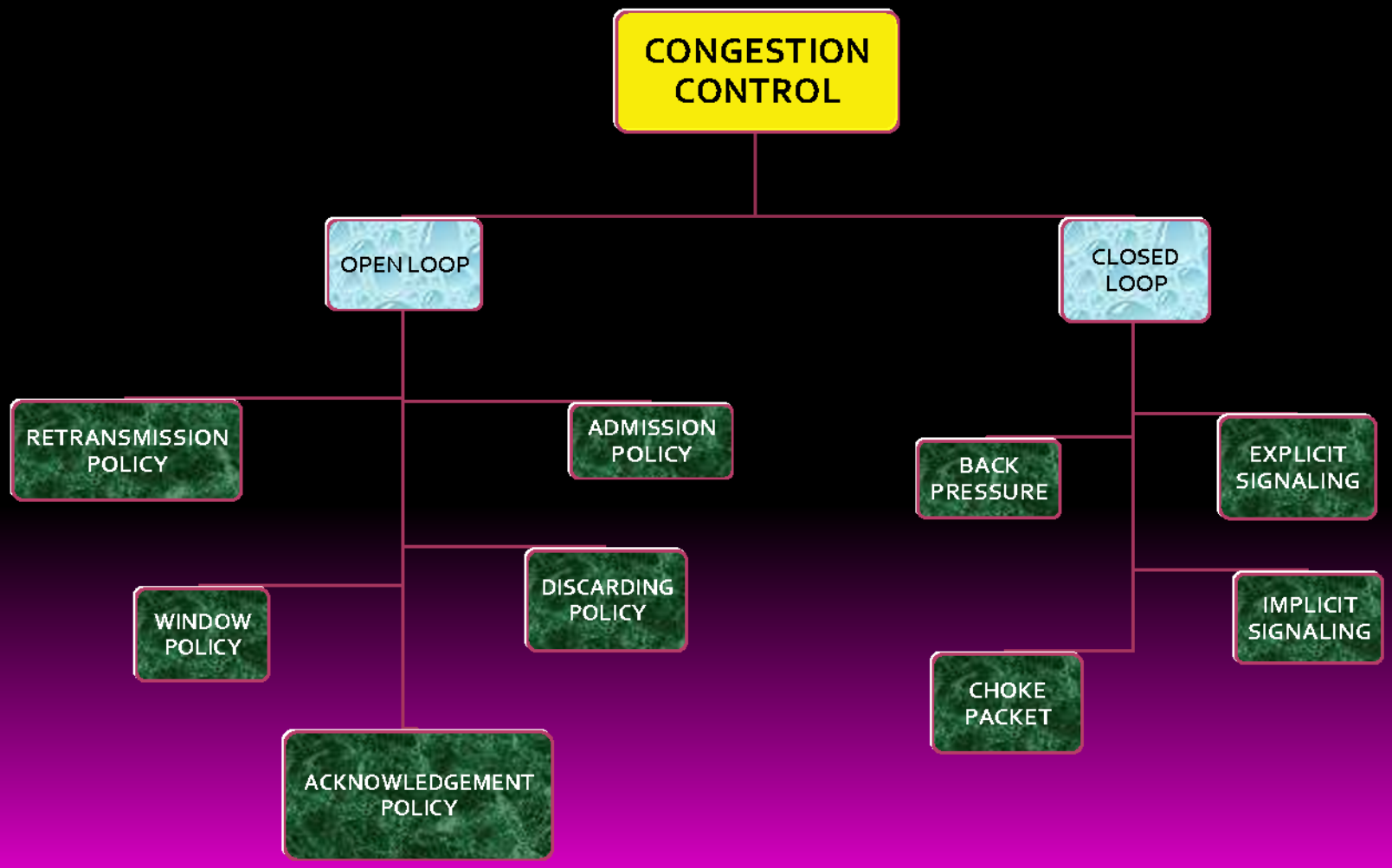
- Packet arrival rate exceeds the outgoing link capacity.
- Insufficient memory to store arriving packets
- Bursty traffic
- Slow processor

Costs of congestion

- large queuing delays are experienced as the packet arrival rate nears the link capacity.
- unneeded retransmissions by the sender
- when a packet is dropped along a path, the transmission capacity of the upstream routers have been wasted.

GENERAL PRINCIPLE OF CONGESTION CONTROL

Congestion control refers to techniques and mechanisms that can either prevent congestion , before it happens or removes congestion,after it has happened.



Open loop congestion control

- In this method, policies are used to prevent the congestion before it happens.

  Congestion control is handled either by the source or by the destination.

1. Retransmission Policy

- The sender retransmits a packet, if it feels that the packet it has sent is lost or corrupted.
- The retransmission policy and the retransmission timers need to be designed to optimize efficiency and at the same time prevent the congestion.

2. Window Policy

- To implement window policy, selective reject window method is used for congestion control in which it sends only the specific lost or damaged packets.

3. Acknowledgement Policy

- If the receiver does not acknowledge every packet it receives it may slow down the sender and help prevent congestion.

4. Discarding Policy

- A router may discard less sensitive packets when congestion is likely to happen

5. Admission Policy

- A router can deny establishing a virtual circuit connection if there is congestion in the "network or if there is a possibility of future congestion.

6. Routing Algorithm

- spreading the traffic over all the lines

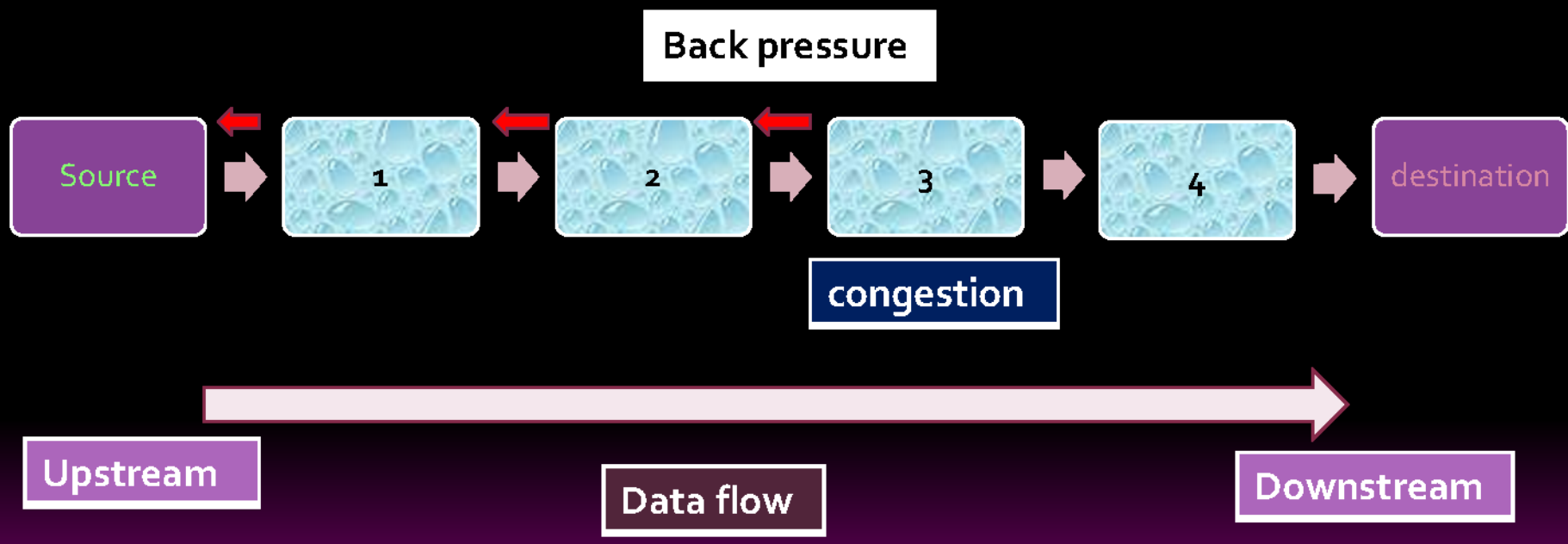
Congestion prevention policies in different layers

LAYER	CONGESTION PREVENTION POLICIES
Transport	-retransmission policy -acknowledgement policy -flow control policy -timeout determination
Network	-virtual circuits versus datagram inside the subnet -packet queuing and service policy -packet discard policy -routing algorithm -packet lifetime management
Datalink	-retransmission policy -out-of-order caching policy -acknowledgement policy -flow control policy

CLOSED LOOP CONGESTION CONTROL

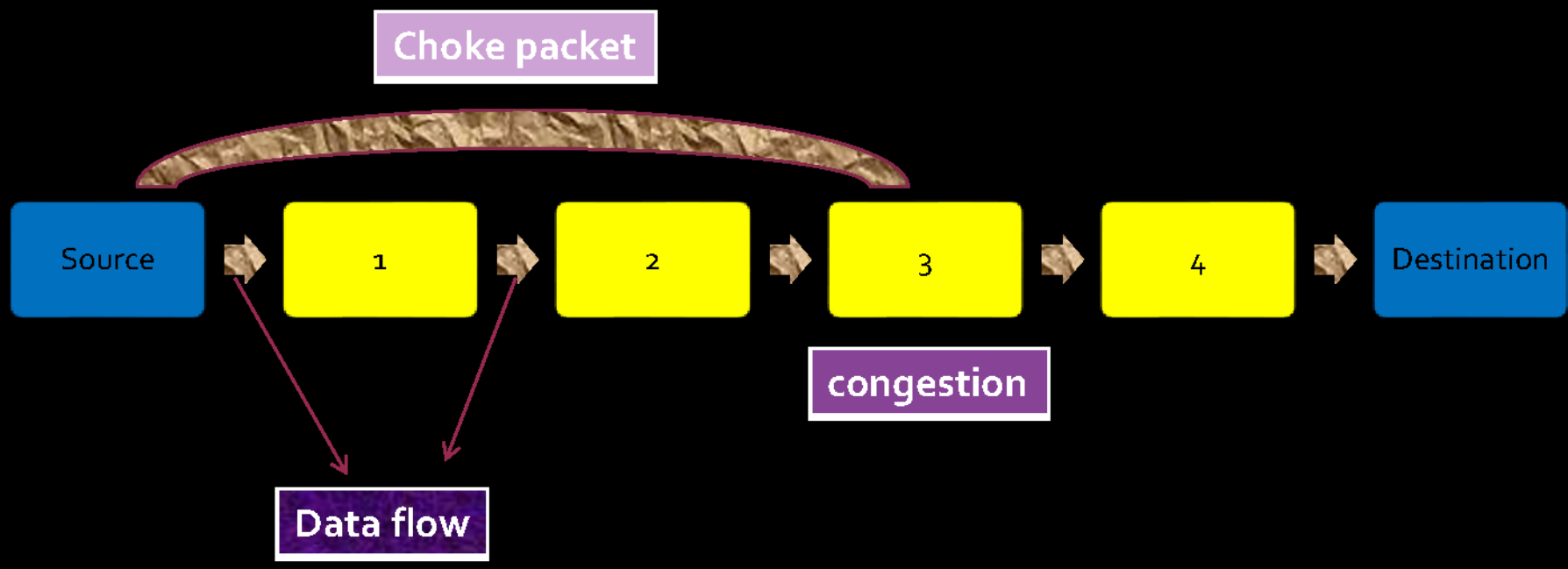
- Closed loop congestion control mechanisms try to remove the congestion after it happens.

1. Backpressure method



- Backpressure is a node-to-node congestion control that starts with a node and propagates, in the opposite direction of data flow in which the congested node stops receiving data from the immediate upstream node(s).

2. Choke Packet



- In choke packet method, congested node sends a warning directly to the source station *i.e.* the intermediate nodes through which the packet has traveled are not warned.

3. Implicit Signaling

- The source guesses that there is congestion somewhere in the network when it does not receive any acknowledgment. Therefore the delay in receiving an acknowledgment is interpreted as congestion in the network and the source slows down.

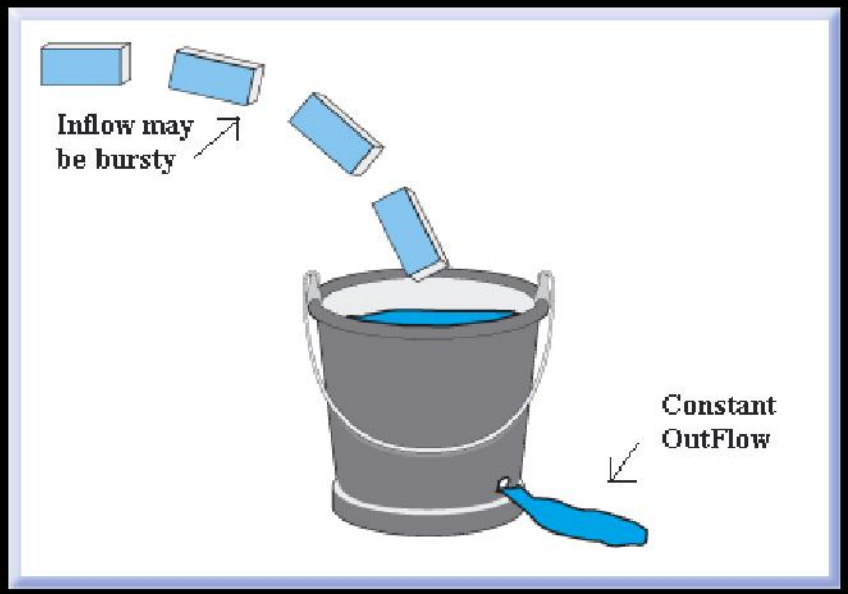
- this policy is used in TCP(Transmission control protocol).

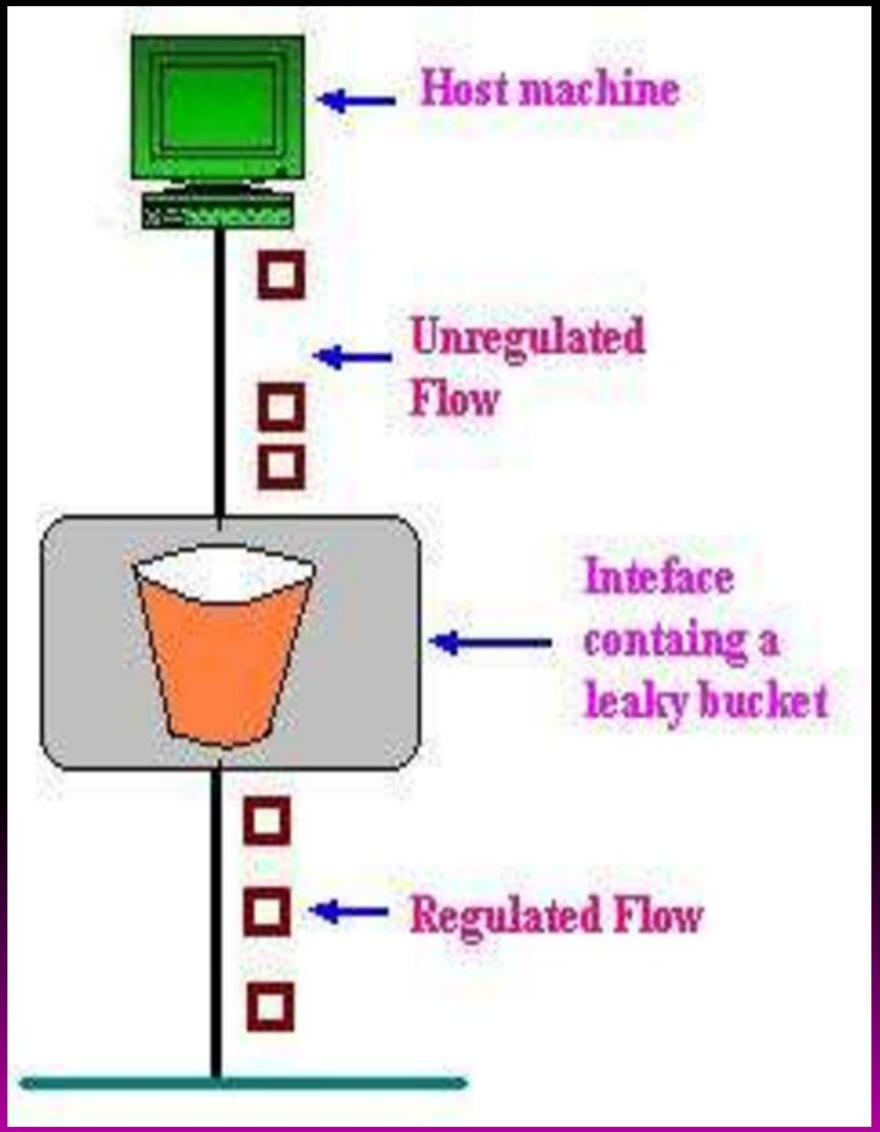
4. Explicit Signaling

- In this method, the congested nodes explicitly send a signal to the source or destination to inform about the congestion.

- Explicit signaling is different from the choke packet method. In choke packed method, a separate packet is used for this purpose whereas in explicit signaling method, the signal is included in the packets that carry data .

LEAKY BUCKET ALGORITHM





TOKEN BUCKET ALGORITHM

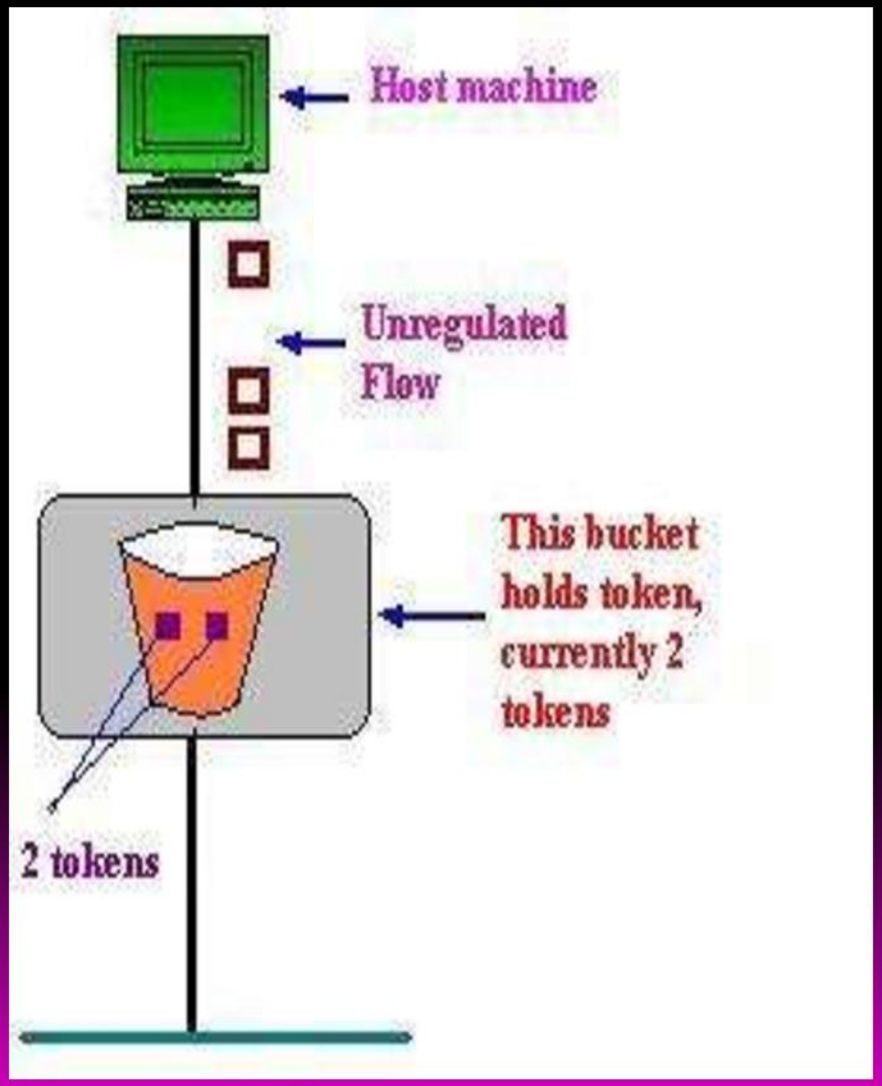


Fig. 1

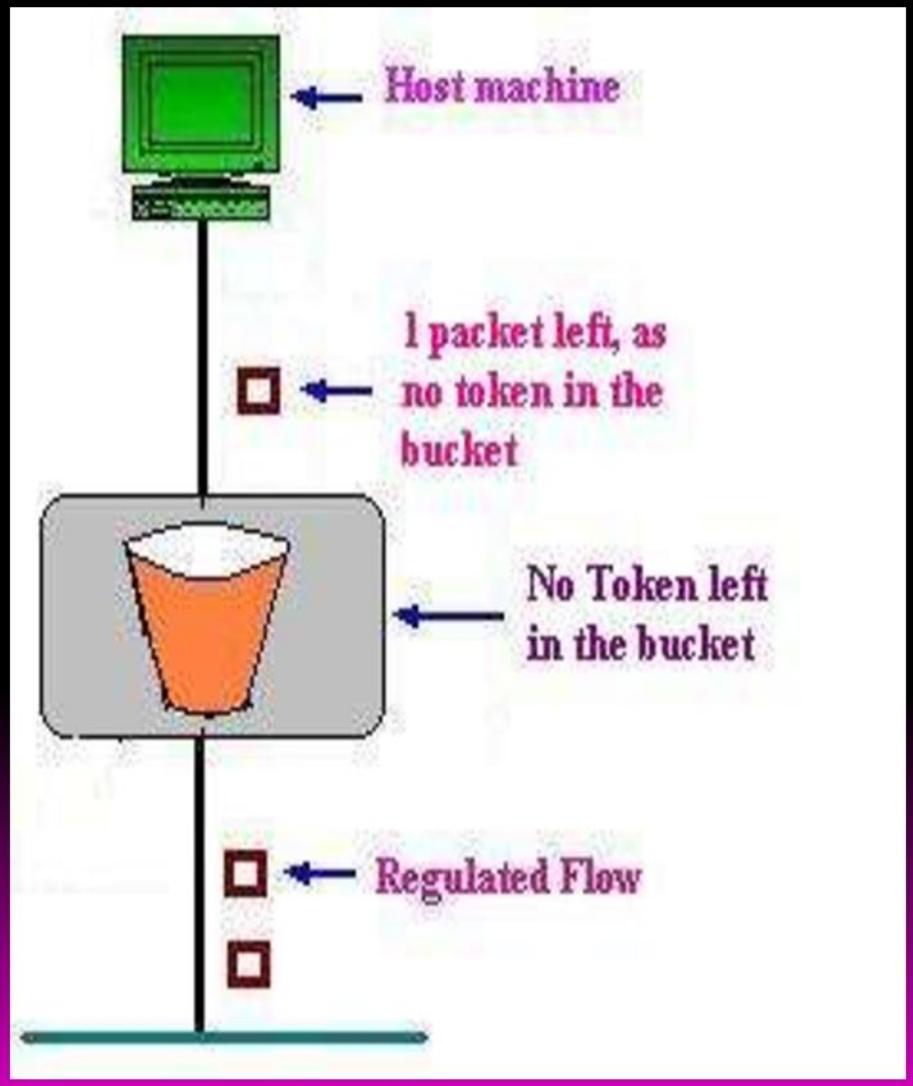
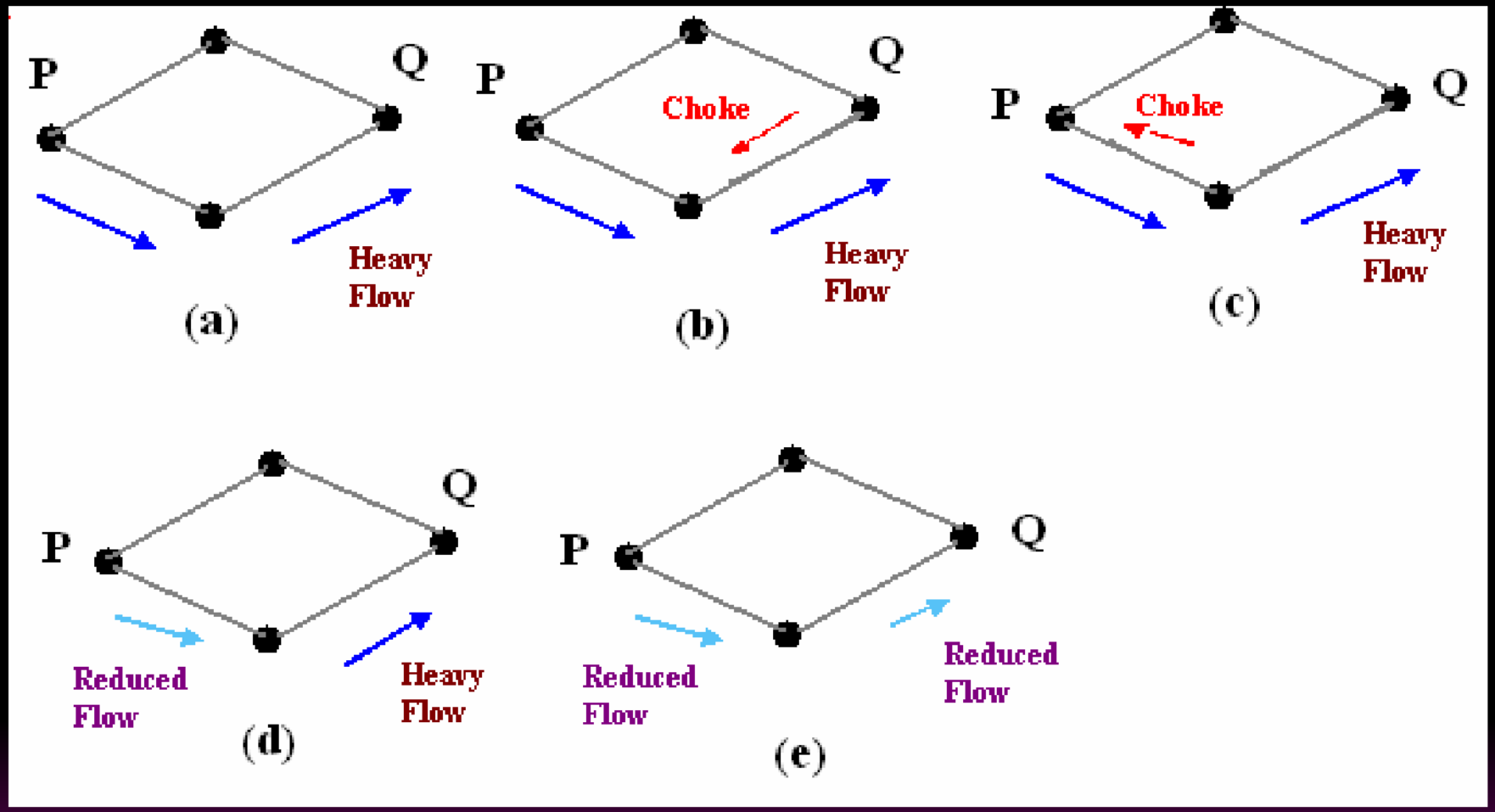


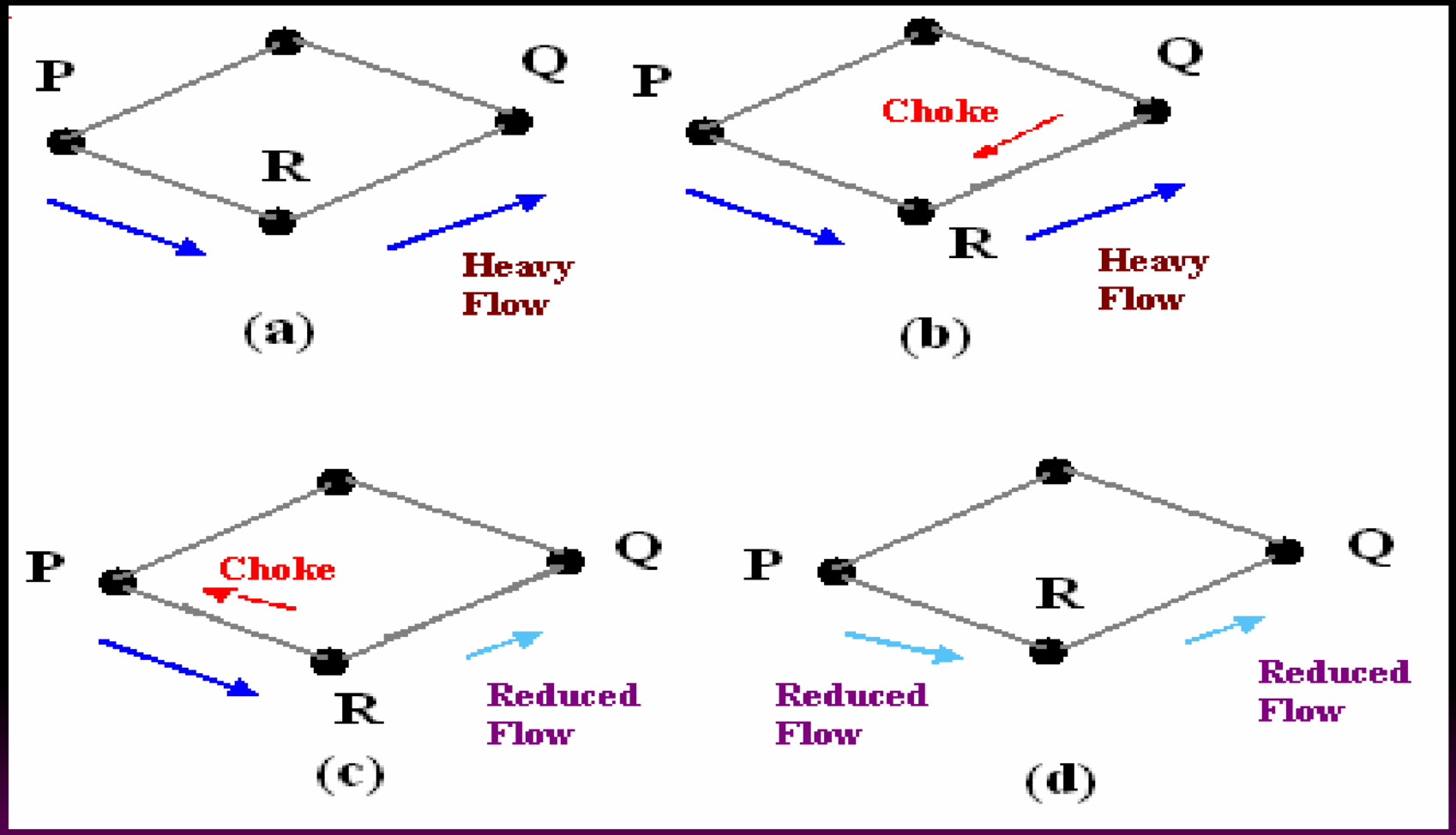
Fig. 2

❖ Choke Packet Technique



❖ Depicts the functioning of choke packets, (a) Heavy traffic between nodes P and Q, (b) Node Q sends the Choke packet to P, (c) Choke packet reaches P, (d) P reduces the flow and send a reduced flow out, (e) Reduced flow reaches node Q .

Hop-by Hop Choke Packets



Depicts the functioning of Hop-by-Hop choke packets (a) Heavy traffic between nodes P and Q, (b) Node Q sends the Choke packet to P, (c) Choke packet reaches R, and the flow between R and Q is curtail down, Choke packer reaches P, and P reduces the flow out .

❖Load Shedding

- **In this method, whenever a router finds that there is congestion in the network, it simply starts dropping out the packets.**
- Which packet to discard depends on the applications running
  - Wine policy: the older the better.
  - Milk policy: fresher is better
  - intelligent discard policy:packets in priority classes

conclusion

- With the increase in the number of internet users, it has become quite essential to develop an efficient congestion control technique.
- In our presentation we have analyse Leaky bucket algorithm, Token bucket algorithm, Choked packet technique and Hop by hop choke packet technique.
- A particular algorithm cannot be hundred percent efficient, so there is always scope to improve the congestion control technique.

Reference

- Computer control overview written by S. Chen
- "What is Congestion Control Describe the Congestion Control Algorithm commonly used." written by Dinesh Thakur.
- Computer Network:A system aproach,5E by Larry L. Peterson & Bruce S. Davie
- **[Ramakrishnan 1990]** K. K. Ramakrishnan and Raj Jain, "A Binary Feedback Scheme for Congestion Avoidance in Computer Networks", *ACM Transactions on Computer Systems*, Vol.8, No.2, pp. 158-181, May 1990.